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Convergence analysis of sparse TSK fuzzy systems based on spectral Dai-Yuan conjugate gradient and application to high-dimensional feature selection[☆]

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ABSTRACT

Dealing with high-dimensional problems has always been a key and challenging issue in the field of fuzzy systems. Traditional Takagi-Sugeno-Kang (TSK) fuzzy systems face the challenges of the curse of dimensionality and computational complexity when applied to high-dimensional data. To overcome these challenges, this paper proposes a novel approach for optimizing TSK fuzzy systems by integrating the spectral Dai-Yuan conjugate gradient (SDYCG) algorithm and the smoothing group L_0 regularization technique. This method aims to address the challenges faced by TSK fuzzy systems in handling high-dimensional problems. The smoothing group L_0 regularization technique is employed to introduce sparsity, select relevant features, and improve the generalization ability of the model. The SDYCG algorithm effectively accelerates convergence and enhances the learning performance of the network. Furthermore, we prove the weak convergence and strong convergence of the new algorithm under the strong Wolfe criterion, which means that the gradient norm of the error function with respect to the weight vector converges to zero, and the weight sequence approaches a fixed point.

1. Introduction

Fuzzy systems, as one of the most promising models, theoretically have the ability to approximate any continuous function with arbitrary precision. Takagi-Sugeno-Kang (TSK) fuzzy systems (Sugeno & Kang, 1986; Takagi & Sugeno, 1985), also known as TSK fuzzy neural networks, are one of the two classical fuzzy systems (the other being the Mamdani fuzzy system (Mamdani & Assilian, 1975)) that have been successfully applied in control engineering, image processing, and other fields. In general, the gradient-based backpropagation algorithm is a popular approach for optimizing TSK fuzzy systems.

The literatures (Fan, Zhang, & Huang, 2022; Gao et al., 2018, 2019; Zhang et al., 2022) propose the use of the conjugate gradient (CG) methods instead of the gradient descent method to optimize the TSK fuzzy systems. Compared with gradient descent method, the CG methods utilize previous gradient information to intelligently select the search direction, resulting in faster convergence to the optimal solution. Additionally, the CG methods do not need to store complete gradient information, which helps to save memory resources and makes them more efficient for large-scale problems.

The conjugate gradient method was originally proposed by Hestenes and Stiefel (Hestenes & Stiefel, 1952) as a technique for solving linear systems of equations, and thus it is also known as the linear conjugate gradient method. Subsequently, Fletcher and Reeves (Fletcher, 1964) applied the conjugate gradient method to solve unconstrained optimization problems, which is often considered the starting point of nonlinear conjugate gradient methods for unconstrained optimization. In addition to these, there are many classical conjugate gradient methods, such as the Polak-Ribière-Polyak (PRP) conjugate gradient method (Polak & Ribiere, 1969), conjugate descent (CD) conjugate gradient method (Fletcher, 1986), Liu-Storey (LS) conjugate gradient method (Liu & Storey, 1991), and Dai-Yuan (DY) conjugate gradient method (Dai & Yuan, 1999). The DY conjugate gradient method exhibits strong convergence, but its experimental performance is sometimes unsatisfactory. Consequently, in order to fully utilize the advantages of the DY conjugate gradient method and improve numerical performance, we have employed a new spectral conjugate gradient method to construct a CG-based TSK fuzzy system (Zhou & Ni, 2015).

During the training process of a neural network model, the step size (learning rate) plays a crucial role in determining the convergence

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speed. Step size search methods can be divided into exact line search and inexact line search methods. While exact line search can determine the optimal step size for the current iteration, it also comes with higher computational costs. Therefore, in the step size search process, inexact line search is more favored. Commonly used inexact line search methods include the Armijo line search technique (Armijo, 1966), Goldstein line search technique (Nocedal & Wright, 2006), and Wolfe line search technique (Wolfe, 1969). The Wolfe line search technique may suffer from issues such as inaccurate step size selection and getting trapped in local optima. In this paper, we employ the strong Wolfe line search technique to compute the step size, which effectively avoids falling into local optima and enhances global convergence.

In addition to the choice of training algorithm, TSK fuzzy systems also face the “curse of dimensionality” problem when the number of input features is too large. In Cui, Wu, and Xu (2021), the authors discuss the reasons for the failure of TSK fuzzy systems with Gaussian membership functions (MFs) on high-dimensional inputs and propose two approaches, LogTSK and HTSK, to avoid saturation in the defuzzification process. In Xue, Chang, Wang, Zhang, and Pal (2023), Xue et al. propose an adaptive softmin (Ada-softmin) technique that effectively prevents the occurrence of “numerical underflow” (Cui, Wu, & Huang, 2020) when computing the strength triggered by the product T-norm. Meanwhile, excessive features often increase the complexity of the model and decrease its generalization ability. Addressing the issue of TSK fuzzy systems struggling with high-dimensional datasets becomes particularly important.

Unlike ordinary datasets, high-dimensional datasets typically contain a significant amount of noise and redundant features, which can negatively impact the model and increase computational costs. Feature selection has become an effective approach for handling high-dimensional problems. By selecting the most relevant or predictive subset of features from the original feature set, it can significantly reduce computational complexity, improve model generalization, and enhance interpretability. Importantly, feature selection can address the “curse of dimensionality” issue in TSK fuzzy systems. There are typically three methods of feature selection: filter methods (Solorio-Fernández, Fco. Martínez-Trinidad, & Carrasco-Ochoa, 2017), wrapper methods (Kohavi & John, 1997), and embedded methods (Chen, Pal, & Chung, 2012). Embedded methods, which integrate feature selection with the model training process, simultaneously perform feature selection and evaluation, making them an increasingly attractive topic (Xue, Chang, et al., 2023; Xue, Wang, Yuan, & Dai, 2023).

Sparse-induced feature selection (FS) methods are new embedded/integrated approaches proposed by researchers in recent years. They incorporate sparse-induced regularization techniques into network models to achieve integrated feature selection. One popular regularization method is L_p regularization, which adds the L_p norm of network weights as a regularization term to the loss function during training. When p equals 2, L_2 regularization is the most common weight decay regularization method, but it lacks sparsity and cannot remove redundant weights. Only when p is less than or equal to 1 can L_p regularization methods generate sparse solutions. L_1 regularization and $L_{1/2}$ regularization have been used to prune network structures and remove redundant weights (Liu & Yang, 2017; Wu et al., 2014; Zhao et al., 2024). In Gao, Bai, Zhang, and Wang (2021), Wang, Zhang, Wang, Pu, and Pal (2021), the authors combine the L_2 norm and L_1 norm to form the $L_{2,1}$ norm, which is then used as a Group Lasso penalty for feature selection. In Kang, Fan, Zurada, and Huang (2022), the sigma-pi-sigma neural network is pruned using group $L_{1/2}$ regularization technique. Sparse regularization-based group L_p regularization methods have become hot topics (An, Dong, & Qin, 2021; Fan, Kang, Zurada, Huang, & Xu, 2024; Fan et al., 2023).

Due to the fact that L_0 regularization can generate the sparsest solution among all L_p regularization methods, we aim to use group L_0 regularization for feature selection. However, L_0 regularization is non-differentiable and solving group L_0 regularization is an NP-hard

problem. To address this, we employ a smooth function sequence to approximate the group L_0 regularization term (Zhang & Tang, 2017; Zhang, Tang, & Liu, 2014; Zhang, Wei, Xu, & Zhang, 2022) and propose a spectral Dai-Yuan conjugate gradient (SDYCG) algorithm with smoothing group L_0 regularization to optimize the TSK fuzzy systems. The main contributions of this paper are as follows:

(1) In this paper, we replaced the traditional error function with the cross-entropy loss function. This is mainly because, in multi-classification tasks, the cross-entropy loss function performs better in handling differences in probability distributions, thereby more effectively guiding the training of the model. Additionally, the cross-entropy loss function can directly optimize the logarithmic loss of classification probabilities, usually leading to faster model convergence and more efficient parameter adjustment compared to the mean squared error function, with better numerical performance.

(2) In the TSK fuzzy system, we consider for the first time the use of the DY conjugate gradient method to optimize the cross-entropy loss function. Based on the DY conjugate gradient method, we introduce spectral parameters to form the SDYCG algorithm. This algorithm reduces training time and improves numerical performance. More importantly, under the strong Wolfe conditions, we prove the weak and strong convergence of the algorithm, effectively enriching and advancing the theoretical results of the TSK fuzzy system.

(3) The cross-entropy loss function considers the group L_0 regularization method, which effectively sparsifies and optimizes the network structure. The effectiveness of the algorithm has been verified through extensive numerical experiments.

The remainder of this article is organized as follows: In Section 2, we describe the TSK fuzzy system and SDYCG algorithm with smoothing group L_0 regularization. Section 3 provides the convergence results of the algorithm, and the corresponding proofs are provided in the appendix. The numerical experiments in Section 4 validate the effectiveness of the proposed algorithm. In the final section, several conclusions are drawn.

2. Training algorithm for TSK

2.1. TSK fuzzy systems

Assuming $\{x_n, y_n\}_{n=1}^N$ is a training dataset, where $x_n = [x_{n,1}, \dots, x_{n,D}]^T \in \mathbb{R}^{D \times 1}$ is the n th input sample with D feature vectors, and $y_n \in \{1, 2, \dots, C\}$ is the corresponding class label in a C -class classification problem.

For a C -class classification problem with input dimension of D and R rules, the r -th fuzzy rule of a first-order TSK model is described as follows:

Rule _{r} : IF x_1 is $X_{r,1}$ and $\dots$ and x_D is $X_{r,D}$,

$$\begin{aligned} \text{THEN } y_r^1(x) &= p_{r,0}^1 + \sum_{d=1}^D p_{r,d}^1 x_d, \dots, \\ y_r^C(x) &= p_{r,0}^C + \sum_{d=1}^D p_{r,d}^C x_d, \end{aligned} \quad (2.1)$$

where $X_{r,d}$ ($r = 1, 2, \dots, R$; $d = 1, 2, \dots, D$) is the fuzzy set corresponding to the d th feature in the r -th rule, $y_r^c(x)$ ($c = 1, 2, \dots, C$) is the output of the c th class in the r -th rule, and $p_{r,d}^c$ represents the consequent parameter of the r -th rule associated with the d th feature for the c th class.

Consider using Gaussian function as the membership function. The membership value μ of x_d on $X_{r,d}$ is:

$$\mu_{r,d}(x_d) = \exp\left(-\frac{(x_d - m_{r,d})^2}{2\sigma_{r,d}^2}\right), \quad (2.2)$$

where $m_{r,d}$ and $\sigma_{r,d}$ respectively represent the center and the standard deviation of the fuzzy set $X_{r,d}$. In Wu, Li, Yang, and Liu (2010), the authors used $\mu = \exp(-(x_d - m_{r,d})^2/b_{r,d}^2)$ as membership function, in

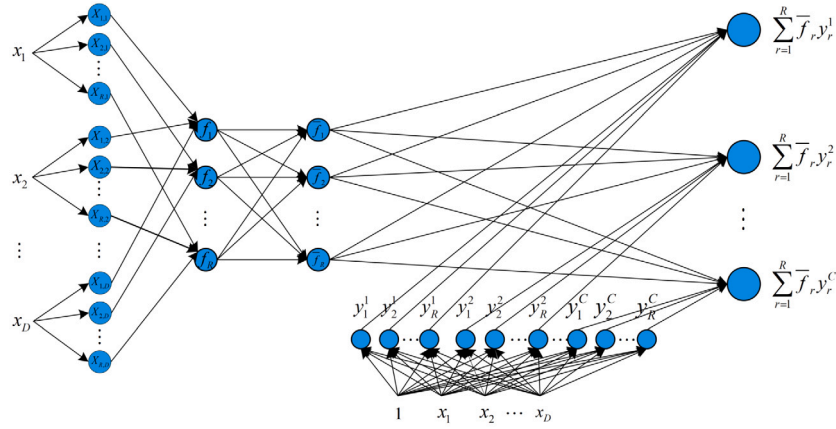


Fig. 1. The topology of a first-order TSK fuzzy system.

which $b_{r,d}$ is the reciprocal of the standard deviation $\sigma_{r,d}$. Similarly, by adopting this denominator-free approach, we can effectively avoid the phenomenon of oscillation during the training process.

The firing strength f_r of the r -th rule can be calculated by the product:

$$f_r(x) = \prod_{d=1}^D \mu_{r,d}(x_d) = \exp\left(-\sum_{d=1}^D (x_d - m_{r,d})^2 b_{r,d}^2\right). \quad (2.3)$$

The c th component of the final output of the TSK fuzzy system is:

$$y^c(x) = \sum_{r=1}^R \bar{f}_r(x) y_r^c(x), \quad (2.4)$$

where

$$\bar{f}_r(x) = \frac{f_r(x)}{\sum_{i=1}^R f_i(x)} \quad (2.5)$$

is the normalized firing strength of the r -th rule.

For clarity, the network structure of a first-order TSK fuzzy system is shown in Fig. 1. For classification problems, the model output is typically a probability distribution that represents the probabilities of each class. The cross-entropy function compares the similarity between two probability distributions and is more suitable for measuring the loss in classification problems. Therefore, the cross-entropy loss function can be used as the error function for multi-class problems (Cui et al., 2020):

$$\bar{E}(w) = -\frac{1}{N} \sum_{n=1}^N \sum_{c=1}^C I(y_n == c) \log(q^c(x_n)), \quad (2.6)$$

where

$$q^c(x_n) = \frac{\exp(y^c(x_n))}{\sum_{c=1}^C \exp(y^c(x_n))}, \quad (2.7)$$

and $I(y_n == c)$ is indicator functions, If the true class of the sample n equals c , take 1, otherwise take 0.

The gradients of the error function with respect to the preceding parameters and the subsequent parameters are as follows:

$$\frac{\partial \bar{E}(w)}{\partial m_{r,d}} = 2 \frac{1}{N} \sum_{n=1}^N \sum_{c=1}^C [q^c(x_n) - I(y_n == c)] \times (y_r^c(x_n) - y^c(x_n)) \bar{f}_r(x_n) (x_{n,d} - m_{r,d}) b_{r,d}^2, \quad (2.8)$$

$$\frac{\partial \bar{E}(w)}{\partial b_{r,d}} = -2 \frac{1}{N} \sum_{n=1}^N \sum_{c=1}^C [q^c(x_n) - I(y_n == c)] \times (y_r^c(x_n) - y^c(x_n)) \bar{f}_r(x_n) (x_{n,d} - m_{r,d})^2 b_{r,d}, \quad (2.9)$$

$$\frac{\partial \bar{E}(w)}{\partial p_{r,d}^c} = \frac{1}{N} \sum_{n=1}^N [q^c(x_n) - I(y_n == c)] \bar{f}_r(x_n) x_{n,d}, \quad (2.10)$$

where $x_{n,d}$ is the d th component of sample x_n .

2.2. Smoothing group L_0 regularization

Considering regularization methods can improve the generalization ability of the network and select useful features, L_0 regularization, among all regularization methods, can generate the sparsest solutions. In this paper, we adopt an ensemble feature selection method based on group L_0 regularization. We treat the reciprocal of the width corresponding to feature x_d , denoted as $\{b_{1,d}, \dots, b_{R,d}\}$, as a group, and then use the L_2 norm and square operator, i.e., $\|b_d\|^2 = (b_{1,d}^2 + \dots + b_{R,d}^2)$.

To discard irrelevant feature x_d , we require the magnitude of $b_{r,d}$ corresponding to that feature to be within $[0, 0.1]$. It is easy to conclude that when $\|b_d\|^2 > 0.01R$, feature x_d may be considered as good (Gao et al., 2021).

Therefore, the error function with the group L_0 regularization term is defined as follows:

$$E(w) = \bar{E}(w) + \lambda \|b\|_0^0, \quad (2.11)$$

where λ is the regularization coefficient, $\|b\|_0^0$ represents the number of non-zero elements in a vector, and it is defined as follows:

$$\|b\|_0^0 = \lim_{p \rightarrow 0^+} (\|b_1\|^{2p} + \|b_2\|^{2p} + \dots + \|b_D\|^{2p}) \quad (2.12)$$

where $b = (\|b_1\|^2, \|b_2\|^2, \dots, \|b_D\|^2)$.

Since the minimization of (2.11) is a combinatorial optimization problem, it cannot be directly solved using gradient-based methods. To address this issue, we consider using multivariate continuous functions:

$$H_\sigma(b) = \sum_{d=1}^D h_\sigma(\|b_d\|^2) \quad (2.13)$$

to approximate the group L_0 regularization term, where $h_\sigma(\cdot)$ is continuously differentiable on R and satisfies

$$\lim_{\sigma \rightarrow 0} h_\sigma(t) = \begin{cases} 1, & t > 0, \\ 0, & t = 0, \end{cases} \quad (2.14)$$

where the parameter σ is a positive value used to control the degree to which $H_\sigma(b)$ approximates the group L_0 regularization term.

Then, the error function with smoothing group L_0 regularization is as follows:

$$E(w) = \bar{E}(w) + \lambda \sum_{d=1}^D h_\sigma(\|b_d\|^2). \quad (2.15)$$

We can take

$$h_\sigma(t) = 1 - \exp\left(\frac{-t}{2\sigma^2}\right) \quad (2.16)$$

as a smoothing function.

Correspondingly, the gradient of the error function $E(w)$ with respect to the parameter $b_{r,d}$ is:

$$\begin{aligned} \frac{\partial E(w)}{\partial b_{r,d}} = & -2 \frac{1}{N} \sum_{n=1}^N \sum_{c=1}^C [[q^c(x_n) - I(y_n == c)] \\ & \times (y_r^c(x_n) - y^c(x_n)) \bar{f}_r(x_n)(x_{n,d} - m_{r,d})^2 b_{r,d}] \\ & + 2\lambda h'_\sigma(\|b_d\|^2) b_{r,d} \end{aligned} \quad (2.17)$$

2.3. SDY conjugate gradient method

The parameter update formula for a fuzzy system is expressed as:

$$w^{k+1} = w^k + \eta^k d^k, \quad (2.18)$$

where $k(k = 0, 1, 2, \dots)$ indicates the k th iteration, η^k represents the learning rate and d^k is the updating direction in the k th iteration. The updating direction for the spectral conjugate gradient method is as follows:

$$d^k = \begin{cases} -E_w^k, & k = 0, \\ -\theta_k E_w^k + \beta^k d^{k-1}, & k \geq 1, \end{cases} \quad (2.19)$$

where E_w^k represents the gradient of the error function $E(w)$ with respect to the parameter w^k . In this paper, we introduce a new spectral parameter (Zhou & Ni, 2015)

$$\theta_k = \max \left\{ \frac{2|(E_w^{k-1})^T d^{k-1}|}{|(d^{k-1})^T (E_w^k - E_w^{k-1})|}, \frac{2(E_w^k)^T d^{k-1} - (E_w^{k-1})^T d^{k-1}}{(d^{k-1})^T (E_w^k - E_w^{k-1})} \right\}, \text{ and}$$

$\beta^k = \frac{\|E_w^k\|^2}{(d^{k-1})^T (E_w^k - E_w^{k-1})}$ is the classical DY conjugate coefficient.

We assume that the step size satisfies the strong Wolfe conditions, that is, the step size η^k satisfies:

$$E(w^k + \eta^k d^k) - E(w^k) \leq \delta \eta^k (E_w^k)^T d^k, \quad (2.20)$$

$$|E_w(w^k + \eta^k d^k)^T d^k| \leq \sigma |(E_w^k)^T d^k|, \quad (2.21)$$

where $0 < \delta < \sigma < 1$.

The pseudocode of SDYCG algorithm is shown in Algorithm 1.

Algorithm 1: Training procedure of the SDYCG.

Input: Training datasets $\{x_n, y_n\}_{n=1}^N$, the stop error err, the maximum iterations K

Output: Optimal w^*

```

1: Initialize  $w^0$  randomly.
2: if  $\|E_w^0\| \leq \text{err}$  then
3:   return  $w^* = w^0$ .
4: else
5:    $d^0 = -E_w^0$ .
6: end if
7: for  $k = 0$  to  $K$  do
8:   Search the learning rate  $\eta^k$  satisfying the strong Wolfe conditions.
9:   Update the weights  $w^{k+1} = w^k + \eta^k d^k$ .
10:  if  $\|E_w^{k+1}\| \leq \text{err}$  then
11:    return  $w^* = w^{k+1}$ .
12:  else
13:     $k = k + 1$ .
14:  end if
15:  Calculate the updating direction  $d^k = -\theta_k E_w^k + \beta^k d^{k-1}$ .
16:  if  $k == K$  then
17:    return  $w^* = w^K$ .
18:  end if
19: end for
```

3. Convergence results

This section presents the convergence theorems of the learning algorithm, and the corresponding proof process is provided in Appendix. First, the assumptions required for the convergence analysis are given as follows:

(A1) The level set $\Omega = \{w : E(w) \leq E(w_0)\}$ is bounded, where w_0 is the initial point.

Table 1

Summary of the eighteen datasets.

Dataset	Num. of samples	Num. of features	Num. of classes
Ecoli	336	7	8
Pima	768	8	2
Yeast	1484	8	10
Glass	214	9	6
Vowel	984	10	11
Vehicle	846	18	4
German	1000	24	2
Biodeg	1055	41	2
Spambase	4597	57	2
Isolet	1560	617	26
Parkinson	756	753	2
CNAE-9	1080	856	9
QSAR androgen receptor	1687	1024	2
QSAR oral toxicity	8992	1024	2
Colon	62	2000	2
Gisette	7000	5000	2
DrivFace	606	6400	3
REJAFADA	1996	6824	2

(A2) The error function $E(w)$ is continuously differentiable and its gradient $E_w(w)$ is Lipschitz continuous in the neighborhood N of the level set Ω .

(A3) In the training process, $\|w^k\| (k = 1, 2, \dots)$ is uniformly bounded.

(A4) There exists a closed bounded region ϕ such that $\{w^k\}_{k=0}^\infty \subset \phi$, and the set $\phi_0 = \{w \in \phi : E_w(w) = 0\}$ contains finite points.

Theorem 3.1.

Assuming the error function $E(w)$ satisfies assumptions (A1)–(A3), and the weight sequence $\{w^k\}$ is updated by SDYCG, we can obtain

$$E(w^{k+1}) \leq E(w^k), \quad k = 0, 1, 2, \dots \quad (3.1)$$

Theorem 3.2. Suppose that the assumptions (A1)–(A3) are valid, then the weight sequence $\{w^k\}$ that is generated by SDYCG satisfies the following weak convergence:

$$\lim_{k \rightarrow \infty} \|E_w^k\| = 0, \quad k \in \mathbb{N}. \quad (3.2)$$

Theorem 3.3. If the assumptions (A1)–(A4) are valid, the following strong convergence results can be obtained:

$$\lim_{k \rightarrow \infty} w^k = w^*, \quad k \in \mathbb{N}, \quad (3.3)$$

where $w^* \in \phi_0$ is the optimal solution.

4. Numerical experiments

In this section, we selected eighteen classification datasets, consisting of nine low-dimensional datasets and nine high-dimensional datasets, to demonstrate the effectiveness of the proposed model. In our study, datasets with more than 500 features were classified as high-dimensional datasets. Table 1 provides a comprehensive summary of these datasets, including the number of samples, the number of features, and the number of classifications.

4.1. Experimental settings

For our experiments, all datasets were randomly divided into training and testing sets with a ratio of 7 : 3. Both the training and testing samples were preprocessed using normalization techniques. The parameter m in the antecedent part of the fuzzy system was obtained through FCM clustering, while the reciprocal of the width, b , was initialized around 0.1. The loss function employed was the cross-entropy loss, and the experimental results were the average of 10 trials. The algorithm stops iterating when either of the following two conditions is met: the norm of the gradient is less than $1e-6$, or the maximum number of iterations is reached.

Table 2

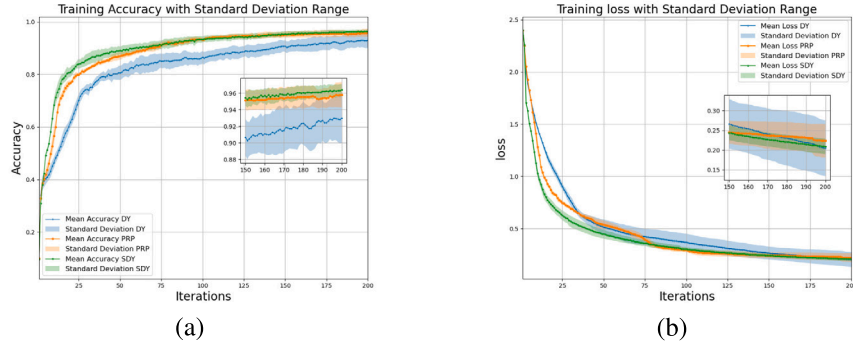
Performance comparison for classification problems with 6 rules.

Dataset	Iterations	DYCG			PRPCG			SDYCG		
		Training time (s)	Training accuracy	Test accuracy	Training time (s)	Training accuracy	Test accuracy	Training time (s)	Training accuracy	Test accuracy
Ecoli	400	41.3762	0.9443	0.8198	44.6848	0.9570	0.8198	16.7755	0.9651	0.8297
Pima	300	29.2748	0.8071	0.7238	35.1310	0.8950	0.7299	17.6259	0.9006	0.7169
Yeast	500	58.4540	0.6618	0.5675	60.0303	0.6908	0.5617	37.5011	0.6939	0.5693
Glass	600	50.6038	0.9409	0.6785	63.1251	0.9685	0.6554	25.4425	0.9483	0.6877
Vowel	300	30.6605	0.9597	0.8439	51.6037	0.9733	0.8351	25.4401	0.9763	0.8591
Vehicle	300	47.4810	0.9530	0.8012	72.8694	0.9380	0.7992	36.1473	0.9350	0.8091
German	500	64.7122	0.8554	0.6847	63.2578	0.9863	0.6787	36.3404	0.9843	0.6940
Biodeg	300	55.3632	0.9509	0.8366	35.0394	0.9864	0.8284	24.1804	0.9886	0.8448
Spambase	200	334.0391	0.9680	0.9196	105.5914	0.9723	0.9305	53.0899	0.9726	0.9341

Table 3

Performance comparison for classification problems with 10 rules.

Dataset	Iterations	DYCG			PRPCG			SDYCG		
		Training time (s)	Training accuracy	Test accuracy	Training time (s)	Training accuracy	Test accuracy	Training time (s)	Training accuracy	Test accuracy
Ecoli	400	50.5421	0.9515	0.7802	68.6599	0.9749	0.8038	30.9844	0.9791	0.8109
Pima	300	28.1814	0.8086	0.7403	23.5838	0.9326	0.7229	14.5382	0.9449	0.7152
Yeast	500	87.2119	0.6678	0.5776	84.9165	0.7328	0.5561	61.6004	0.7362	0.5574
Glass	600	63.5165	0.9470	0.6508	84.0419	0.9893	0.6662	32.4043	0.9725	0.6800
Vowel	300	53.8198	0.9672	0.8517	38.5820	0.9887	0.8635	15.2858	0.9924	0.8760
Vehicle	300	57.3234	0.9645	0.7839	75.8948	0.9688	0.7811	46.7670	0.9552	0.7980
German	500	66.6184	0.8686	0.6867	14.6523	1.0000	0.6847	14.3580	1.0000	0.6787
Biodeg	300	137.5880	0.9553	0.8315	89.7013	0.9897	0.8385	58.9617	0.9911	0.8385
Spambase	200	427.2301	0.9618	0.9225	177.6766	0.9742	0.9261	142.0013	0.9826	0.9363

**Fig. 2.** Comparison of training accuracy and training loss with standard deviation range on the Vowel dataset.

4.2. Performance comparison

We first conducted a numerical comparison between the proposed new algorithm SDYCG and the classical conjugate gradient algorithms Dai-Yuan conjugate gradient (DYCG) and Polak-Ribière-Polyak conjugate gradient (PRPCG) on nine low-dimensional datasets.

Tables 2 and 3 display the training time, training accuracy, and testing accuracy of the three algorithms under different rules of the fuzzy system. The results indicate that the SDYCG algorithm achieves higher training accuracy and testing accuracy than the DYCG algorithm on most datasets, and the proposed algorithm performs comparably well to the PRPCG algorithm, which is known for its superior numerical performance. Furthermore, on all datasets, the new algorithm has a shorter training time compared to other algorithms. This is because our algorithm possesses a faster descent direction, making it easier to search for step sizes using the strong Wolfe condition.

To clearly illustrate the effectiveness of the algorithm, we take the Vowel dataset as an example. Figs. 2(a) and 2(b) depict the training accuracy and training loss curves with standard deviation ranges. The shaded regions represent the standard deviation ranges for the three algorithms. The training accuracy and training loss values are obtained from the average of 10 repeated experiments, and the standard deviation is also calculated based on these 10 repeated experiments.

From Fig. 2(a), we can observe that the DYCG algorithm has a lower average training accuracy and a larger standard deviation range, indicating its poor stability. This observation is further supported by the average training loss plot in Fig. 2(b). On the other hand, the SDYCG algorithm demonstrates faster convergence speed and higher average training accuracy. Additionally, the shaded range in the SDYCG algorithm curve also indicates its stability compared to the other algorithms.

4.3. Feature selection

To perform a comprehensive comparison of various regularization coefficient values (λ) on feature selection, we took measures to ensure consistent initial parameters by fixing the random seed. The initial value of the reciprocal of the width was set at 0.1, and a rule count of $R = 10$ was employed. For feature selection, we set the threshold to $0.01R$, and σ was determined as $\sigma = 1/(k^{0.06})$. Furthermore, we conducted a total of 4000 iterations to obtain accurate and reliable results.

In Table 4, we present the effects of different λ values on the experimental outcomes for nine low-dimensional datasets. This detailed analysis allows us to gain insights into the impact of different regularization settings on feature selection performance. By examining the experimental results, we can determine the optimal λ values that yield

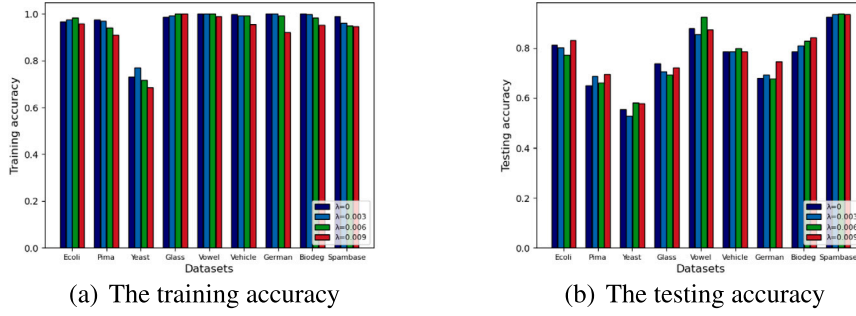
Fig. 3. Training accuracy and testing accuracy under different values of λ .

Table 4
Selection results of different λ on low-dimensional datasets.

Dataset	λ	Training accuracy	Test accuracy	Selected features
Ecoli	0	0.9659	0.8119	7
	0.003	0.9744	0.8020	6
	0.006	0.9829	0.7723	3
	0.009	0.9574	0.8317	4
Pima	0	0.9739	0.6494	8
	0.003	0.9683	0.6883	8
	0.006	0.9404	0.6623	7
	0.009	0.9087	0.6970	7
Yeast	0	0.7321	0.5538	8
	0.003	0.7697	0.5291	8
	0.006	0.7177	0.5807	6
	0.009	0.6859	0.5785	5
Glass	0	0.9865	0.7385	9
	0.003	0.9932	0.7077	7
	0.006	1.0000	0.6923	6
	0.009	1.0000	0.7231	6
Vowel	0	1.0000	0.8784	10
	0.003	1.0000	0.8547	9
	0.006	1.0000	0.9257	6
	0.009	0.9883	0.8750	3
Vehicle	0	0.9966	0.7874	18
	0.003	0.9915	0.7874	14
	0.006	0.9915	0.7992	9
	0.009	0.9543	0.7874	8
German	0	1.0000	0.6800	24
	0.003	1.0000	0.6933	18
	0.006	0.9928	0.6767	13
	0.009	0.9214	0.7467	6
Biodeg	0	1.0000	0.7855	41
	0.003	0.9986	0.8107	9
	0.006	0.9837	0.8297	9
	0.009	0.9525	0.8423	3
Spambase	0	0.9894	0.9254	57
	0.003	0.9618	0.9363	16
	0.006	0.9506	0.9377	7
	0.009	0.9468	0.9356	4

the most favorable outcomes in terms of feature selection accuracy and overall model performance.

From Table 4, it can be observed that in cases where the dimensionality is below 10, the regularization method is capable of selecting features, but it does not yield significantly improved results. This can be attributed to the small number of features and their strong correlation. As the dimensionality increases, the SGL_0 regularization method not only selects meaningful features but also enhances the generalization ability of the model. In Fig. 3, the training accuracy and testing accuracy of different regularization coefficients on various datasets are presented in a more intuitive manner using bar charts.

Figs. 4(a) to 4(d) depict the curves of feature selection with different regularization coefficients (λ) on the Vowel dataset. As shown in the

Table 5
The value of $\|b_d\|^2$ associated with features for the Vowel dataset.

$\ b_d\ ^2$	$\lambda = 0$	$\lambda = 0.003$	$\lambda = 0.006$	$\lambda = 0.009$
F1	32.9160	24.0074	9.8741e + 00	1.2941e−30
F2	103.5822	124.3116	8.7403e + 01	2.6903e + 01
F3	34.9845	101.1371	8.0682e + 01	1.4421e + 02
F4	37.9107	45.3832	1.0238e + 01	1.4329e−08
F5	50.9210	23.0941	1.2663e−01	5.8756e−27
F6	35.0184	33.8879	0.0000e + 00	0.0000e + 00
F7	24.3772	13.9960	0.0000e + 00	3.7802e + 00
F8	19.3633	0.0000	2.1260e−30	0.0000e + 00
F9	48.7286	53.3309	8.9340e−01	2.6007e−32
F10	24.2533	5.0497	6.9675e−04	1.4013e−45

Table 6
Performance of different regularization methods on high-dimensional datasets.

Datasets	TSK- L_2	TSK- L_1	TSK- $SL_{1/2}$	TSK-SGL	TSK-SGL $_{1/2}$	TSK-SGL $_0$
	AccTe	AccTe	AccTe	AccTe	AccTe	AccTe
Isolet	0.9338	0.9284	0.9385	0.9348	0.9380	0.9387
Parkinson	0.8366	0.8449	0.8396	0.8454	0.8502	0.8665
CNAE-9	0.9395	0.9404	0.9472	0.9531	0.9559	0.9470
QSAR androgen receptor	0.8698	0.8830	0.8754	0.8777	0.8767	0.8852
QSAR oral toxicity	0.9191	0.9351	0.9255	0.9251	0.9255	0.9248
Colon	0.8526	0.7474	0.7790	0.7632	0.8526	0.8263
Gisette	0.9687	0.9656	0.9734	0.9737	0.9736	0.9765
DrivFace	0.9692	0.9538	0.9659	0.9670	0.9484	0.9725
REJAFADA	0.9075	0.8688	0.9059	0.9062	0.9257	0.9275
Average acc	0.9108	0.8964	0.9056	0.9051	0.9163	0.9183
Average rank	4.4	4.6	3.8	3.3	2.7	2

figures, initially, we assigned all features as “bad” features, denoted as $b_{r,d} \in [0, 0.1]$. Throughout the training process, if a feature is deemed good or contributes significantly to the outcome, the value of $\|b_d\|^2$ increases. Based on this concept, we selected useful features. Table 5 displays the final values after 4000 iterations.

4.4. Performance comparison on high-dimensional datasets

To demonstrate the effectiveness of our proposed sparse regularization method on high-dimensional datasets, we compared our SGL_0 regularization method with five other regularization methods, including L_2 regularization, L_1 regularization, smoothing $L_{1/2}$ regularization ($SL_{1/2}$), smoothing group lasso regularization (SGL), and smoothing group $L_{1/2}$ regularization ($SGL_{1/2}$). All regularization methods employed the same regularization coefficient $\lambda = 0.002$. In our algorithm, σ was determined as $\sigma = 1/(k^{0.01})$. The experiments were repeated 10 times, and the average performance was reported. Table 6 presents the average test accuracy (AccTe) of the six regularization methods on nine high-dimensional datasets, with our proposed SGL_0 regularization method showing promising results on six high-dimensional datasets.

In addition to comparing the regularization methods, we compared our proposed SDYCG algorithm with SGL_0 regularization to other methods, including random forest (RF) classifier, support vector machine

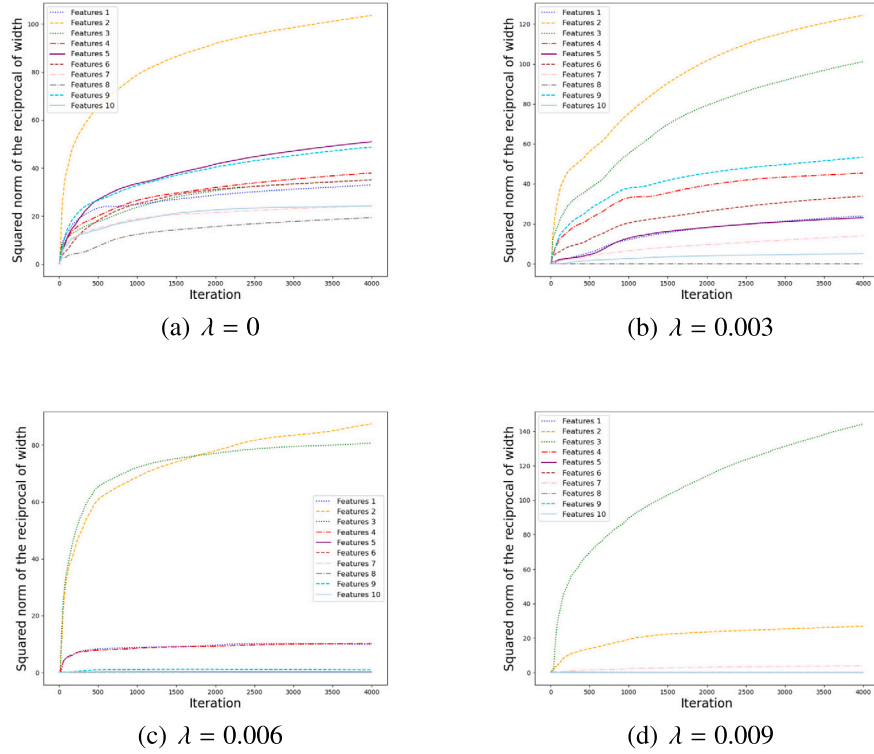


Fig. 4. The squared norm of the reciprocal of the width associated with the features for Vowel.

(SVM) classifier, TSK fuzzy system with feature selection using Pearson correlation coefficient (PCC) and dimensionality reduction using principal component analysis (PCA). These comparisons were conducted to evaluate the performance and superiority of our algorithm in handling high-dimensional datasets.

In RF, the number of trees was set to 100. In PCC, the threshold for the correlation coefficient was set to 0.9. This allowed for the removal of highly correlated features and the selection of features with lower correlation, reducing redundant information. In PCA, the dimensionality was reduced while retaining 60% of the original data. All algorithms were run 10 times, and the average performance was reported. Table 7 presents the classification accuracy of different algorithms, with the best results highlighted in bold.

From Table 7, it can be observed that our algorithm achieves the highest classification accuracy on five out of the nine high-dimensional datasets. Furthermore, compared to any other algorithm, our algorithm only performs relatively poorly on one to two datasets, while demonstrating excellent performance on the majority of high-dimensional datasets. This is evident from the average classification accuracy and average ranking of the six algorithms across the nine high-dimensional datasets. It is clear that our new algorithm is effective and satisfactory in addressing high-dimensional problems.

Figs. 5 presents the top 10 ranked features obtained through feature selection using the SGL_0 algorithm on the nine high-dimensional datasets. The importance of each feature is indicated by the magnitude of $\|b_d\|^2$ values. After training with the SGL_0 algorithm, the $\|b_d\|^2$ values corresponding to important features will increase and be selected. The x -axis represents the index positions of the selected features in the dataset. Through these figures, we can visually and clearly observe the effectiveness of the new algorithm in feature selection.

Meanwhile, to intuitively understand the feature selection effect of our algorithm on high-dimensional datasets, Fig. 6 depicts the variation curve of $\|b_d\|^2$ values on the Isolet dataset. As shown in Fig. 6(a), when the regularization coefficient λ is equal to 0, the algorithm does not have a feature selection effect, and the corresponding $\|b_d\|^2$ values are not suppressed. However, when the regularization coefficient λ is not

Table 7

Performance comparison of different algorithms on high-dimensional datasets.

Datasets	RF	SVM	PCC-TSK	PCA-TSK	TSK-SGL ₀
	AccTe	AccTe	AccTe	AccTe	AccTe
Isolet	0.9205	0.9303	0.9138	0.9145	0.9387
Parkinson	0.8625	0.7498	0.8383	0.8179	0.8665
CNAE-9	0.9052	0.9172	0.9326	0.9228	0.9470
QSAR androgen receptor	0.8860	0.8945	0.8694	0.8671	0.8852
QSAR oral toxicity	0.9432	0.9354	0.9212	0.9294	0.9248
Colon	0.7895	0.8597	0.8684	0.7368	0.8263
Gisette	0.9704	0.9737	0.9695	0.9720	0.9765
DrvFace	0.9576	0.9406	0.9451	0.9049	0.9725
REJAFADA	0.9780	0.9155	0.9392	0.9222	0.9275
Average acc	0.9125	0.9019	0.9108	0.8875	0.9183
Average rank	2.7	3.0	3.3	4.0	2.0

equal to 0, the SGL_0 regularization term plays a role in feature selection. The $\|b_d\|^2$ values corresponding to irrelevant features decrease with increasing iterations, while the $\|b_d\|^2$ values corresponding to relevant features increase, as shown in Fig. 6(b).

5. Conclusion

This paper proposes a novel algorithm for optimizing TSK fuzzy systems. To address the “curse of dimensionality” problem faced by TSK fuzzy systems in high-dimensional settings, we introduce a smoothing group L_0 regularization term, which serves as an embedded feature selection method to select important features during the model training process. Additionally, we introduce a spectral parameter based on the DYCG method, forming the SDYCG algorithm. This algorithm leverages the favorable convergence properties of the DYCG method while addressing its slow convergence speed and subpar performance.

By combining SDYCG with the SGL_0 regularization term for optimizing TSK fuzzy systems, we demonstrate the efficiency of the new algorithm on high-dimensional datasets. Compared to other classical

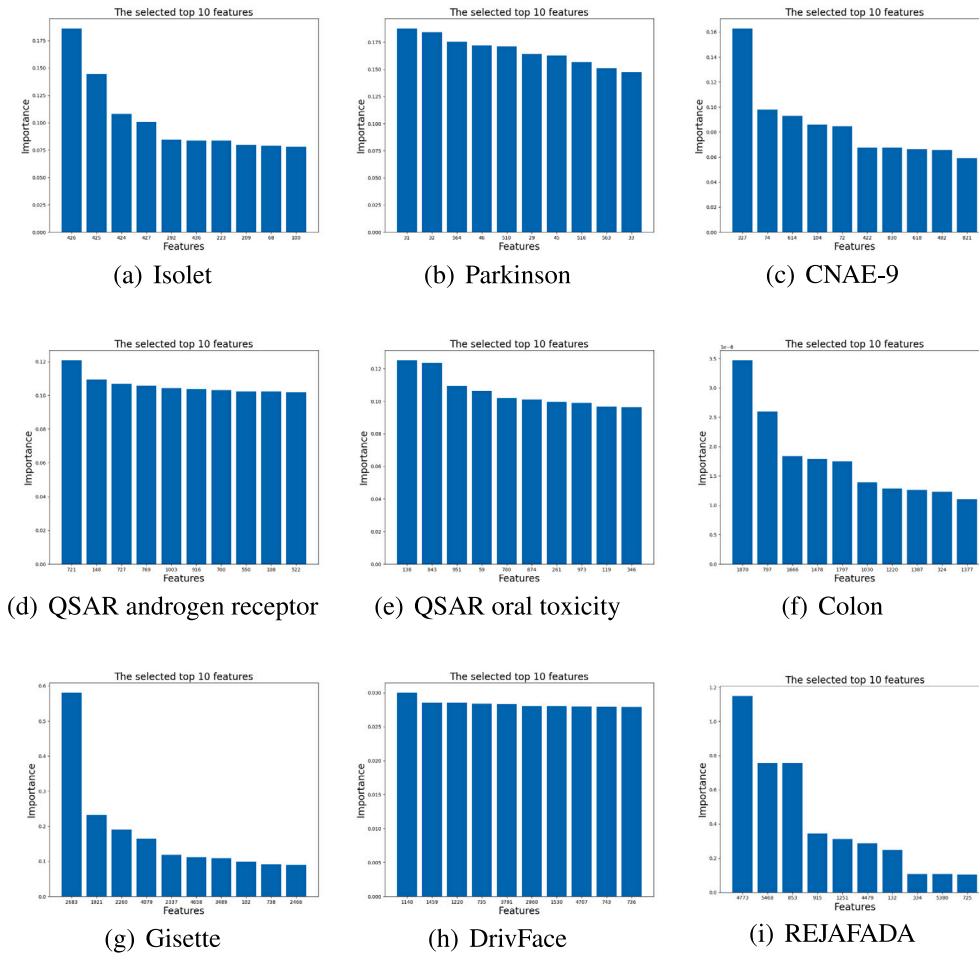


Fig. 5. Top 10 features selected in ten repetitions on high-dimensional datasets.

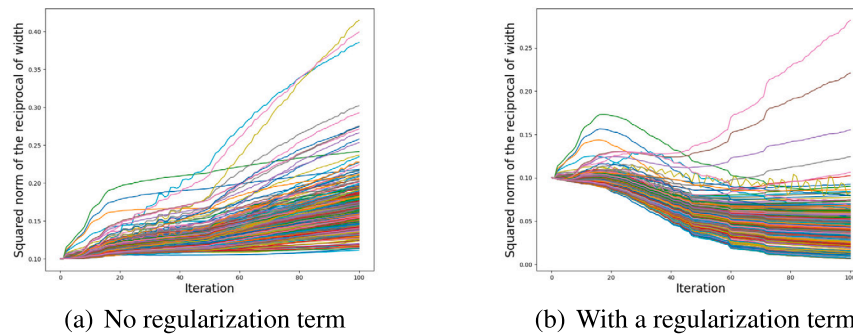


Fig. 6. The squared norm of the reciprocal of the width associated with the features for Isolet.

feature selection algorithms, our algorithm produces satisfactory results. Furthermore, we provide theoretical proofs of the algorithm's convergence. In future research, considering high-dimensional classification problems with tens of thousands of input features will be an important and intriguing task. Feature selection plays a crucial role in machine learning and data mining tasks because it can enhance model performance and reduce computational complexity. Therefore, our algorithm can be applied to various fields such as data mining tasks, while further optimization can improve the predictive accuracy of the model and provide reasonable interpretability. Further exploration of the application of this algorithm in different fields can unlock its potential value.

CRediT authorship contribution statement

Deqing Ji: Writing – original draft, Software. **Qinwei Fan:** Writing – review & editing, Supervision, Methodology, Funding acquisition. **Qingmei Dong:** Software, Data curation. **Yunlong Liu:** Software, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix

The following section provides useful lemmas for the convergence analysis.

Lemma A.1. Assume the weight sequence $\{w^k\}$ is generated by SDYCG algorithm, then the following holds

$$(E_w^k)^T d^k \leq -\|E_w^k\|^2 \quad (\text{A.1})$$

for any $k \geq 0$.

Proof. If $k = 0$, we have

$$(E_w^0)^T d^0 = -\|E_w^0\|^2. \quad (\text{A.2})$$

If $k \geq 1$, from the definition of spectral parameter, we can obtain

$$\theta_k \geq \frac{2(E_w^k)^T d^{k-1} - (E_w^{k-1})^T d^{k-1}}{(d^{k-1})^T (E_w^k - E_w^{k-1})}. \quad (\text{A.3})$$

Furthermore, we can obtain

$$\begin{aligned} (E_w^k)^T d^k &= -\theta_k \|E_w^k\|^2 + \beta^k (E_w^k)^T d^{k-1} \\ &\leq -\left(\frac{2(E_w^k)^T d^{k-1} - (E_w^{k-1})^T d^{k-1}}{(d^{k-1})^T (E_w^k - E_w^{k-1})} \right) \|E_w^k\|^2 \\ &\quad + \beta^k (E_w^k)^T d^{k-1} \\ &\leq \frac{-(E_w^k)^T d^{k-1}}{(d^{k-1})^T (E_w^k - E_w^{k-1})} \|E_w^k\|^2 - \|E_w^k\|^2 \\ &\quad + \beta^k (E_w^k)^T d^{k-1} \\ &\leq -\beta^k (E_w^k)^T d^{k-1} - \|E_w^k\|^2 + \beta^k (E_w^k)^T d^{k-1} \\ &\leq -\|E_w^k\|^2. \end{aligned} \quad (\text{A.4})$$

Thus the conclusion is proved.

Lemma A.2.

Assuming Lemma A.1 holds, then the following inequality holds.

$$\|d^k\| \leq \varepsilon_1 \|E_w^k\| \quad (\text{A.5})$$

Proof. According to (2.19), we can easily obtain

$$\begin{aligned} \|d^k\| &= \|-\theta_k E_w^k + \beta^k d^{k-1}\| \\ &= \|-\theta_k E_w^k + \frac{\|E_w^k\|^2}{(d^{k-1})^T (E_w^k - E_w^{k-1})} d^{k-1}\| \\ &\leq \|\theta_k E_w^k\| + \frac{\|E_w^k\|^2 d^{k-1}}{(d^{k-1})^T (E_w^k - E_w^{k-1})} \\ &\leq \|\theta_k E_w^k\| + \frac{\|E_w^k\|^2 \|d^{k-1}\|}{|(d^{k-1})^T E_w^k|} \\ &\leq (|\theta_k| + \mu) \|E_w^k\| \\ &= \varepsilon_1 \|E_w^k\| \end{aligned} \quad (\text{A.6})$$

where $\mu = \frac{\|E_w^k\| \|d^{k-1}\|}{|(d^{k-1})^T E_w^k|}$, and $\varepsilon_1 = (|\theta_k| + \mu)$.

Lemma A.3.

Assuming that the step size η^k in the SDYCG algorithm satisfies the strong Wolfe line search conditions (2.20) and (2.21), then for any $k \geq 0$, we have:

$$(\beta^k)^2 \leq \frac{((E_w^k)^T d^k)^2}{((E_w^{k-1})^T d^{k-1})^2} \quad (\text{A.7})$$

Proof. Because of $0 < \sigma < 1$, then, according to the definition of spectral parameters, we have

$$\theta_k \geq (\sigma + 1) \frac{|(E_w^{k-1})^T d^{k-1}|}{|(d^{k-1})^T (E_w^k - E_w^{k-1})|} \quad (\text{A.8})$$

By using (2.19) and the second inequality of the strong Wolfe condition (2.21), and then combining it with (A.8), we can obtain

$$\begin{aligned} |(E_w^k)^T d^k| &= |-\theta_k \|E_w^k\|^2 + \beta^k (E_w^k)^T d^{k-1}| \\ &\geq \theta_k \|E_w^k\|^2 - |\beta^k (E_w^k)^T d^{k-1}| \\ &\geq \theta_k \|E_w^k\|^2 - \sigma |\beta^k (E_w^{k-1})^T d^{k-1}| \\ &\geq (\sigma + 1) \frac{|(E_w^{k-1})^T d^{k-1}|}{|(d^{k-1})^T (E_w^k - E_w^{k-1})|} \|E_w^k\|^2 \\ &\quad - \sigma |\beta^k (E_w^{k-1})^T d^{k-1}| \\ &\geq (\sigma + 1) \frac{|\beta^k (E_w^{k-1})^T d^{k-1}|}{\|E_w^k\|^2} \|E_w^k\|^2 \\ &\quad - \sigma |\beta^k (E_w^{k-1})^T d^{k-1}| \\ &\geq |\beta^k (E_w^{k-1})^T d^{k-1}| \end{aligned} \quad (\text{A.9})$$

Thus the conclusion is proved.

Lemma A.4. Assuming that the error function $E(w)$ satisfies assumptions (A1) and (A2), and the search direction d_k is a descent direction, and η_k is obtained by the Wolfe conditions, then

$$\sum_{k=1}^{\infty} \frac{(E_w^T(w^k) d^k)^2}{\|d^k\|^2} < +\infty \quad (\text{A.10})$$

Generally, we refer to the above equation as the Zoutendijk condition. The proof process can be found in Ref. Dai and Yuan (1999), which we omit here.

Lemma A.5. Assuming that the function $F : R^Q \rightarrow R$ is continuous on a bounded closed region $\phi \subset R^Q$, and $\phi_0 = \{x \in \phi : F_w(w) = 0\}$ contains a finite number of points. If the sequence $\{w^k\} \subset \phi$ satisfies:

- (i) $\lim_{k \rightarrow \infty} \|F_w(w^k)\| = 0$,
 - (ii) $\lim_{k \rightarrow \infty} \|w^{k+1} - w^k\| = 0$,
- then there exists a point $x^* \in \phi_0$ such that $\lim_{k \rightarrow \infty} x^k = x^*$.

Proof. The proof process of Lemma A.5 is similar to Theorem 14.1.5 in literature (Yuan & Sun, 2001), so we omit the proof here.

Proving Theorem 3.1. From the first inequality (2.20) of the strong Wolfe condition, we can easily obtain

$$E(w^{k+1}) - E(w^k) \leq \delta \eta^k (E_w^k)^T d^k. \quad (\text{A.11})$$

Then, combining with Lemma A.1, we can obtain

$$(E_w^k)^T d^k \leq 0. \quad (\text{A.12})$$

Therefore,

$$E(w^{k+1}) \leq E(w^k) \quad k = 0, 1, 2, \dots \quad (\text{A.13})$$

The proof of Theorem 3.1 is complete.

Proving Theorem 3.2. The proof is by contradiction. Assuming that Theorem 3.2 does not hold, then there exists a constant $r > 0$ such that

$$\|E_w^k\| \geq r, k = 0, 1, 2, \dots \quad (\text{A.14})$$

From the second equation of (2.19), we can deduce

$$\|d^k\|^2 = (\beta^k)^2 \|d^{k-1}\|^2 - 2\theta_k (E_w^k)^T d^k - \theta_k^2 \|E_w^k\|^2. \quad (\text{A.15})$$

By simultaneously dividing both sides of Eq. (A.15) by $((E_w^k)^T d^k)^2$ and combining it with Eq. (A.7) from Lemma A.3, we can obtain

$$\begin{aligned} & \frac{\|d^k\|^2}{((E_w^k)^T d^k)^2} \\ &= \frac{(\beta^k)^2}{((E_w^k)^T d^k)^2} \|d^{k-1}\|^2 - \left(\frac{1}{\|E_w^k\|} + \frac{\theta_k \|E_w^k\|}{(E_w^k)^T d^k} \right)^2 + \frac{1}{\|E_w^k\|^2} \\ &\leq \frac{\|d^{k-1}\|^2}{((E_w^{k-1})^T d^{k-1})^2} + \frac{1}{\|E_w^k\|^2}. \end{aligned} \quad (\text{A.16})$$

Since $\frac{\|d^0\|^2}{((E_w^0)^T d^0)^2} = \frac{1}{\|E_w^0\|^2}$, by (A.14), we can get

$$\frac{\|d^k\|^2}{((E_w^k)^T d^k)^2} \leq \sum_{i=0}^k \frac{1}{\|E_w^i\|^2} \leq \frac{k+1}{r^2}. \quad (\text{A.17})$$

Then

$$\frac{((E_w^k)^T d^k)^2}{\|d^k\|^2} \geq \frac{r^2}{k+1}, \quad (\text{A.18})$$

hence, we get

$$\sum_{k=0}^{+\infty} \frac{((E_w^k)^T d^k)^2}{\|d^k\|^2} = +\infty. \quad (\text{A.19})$$

If the step size satisfies the strong Wolfe condition, then the Wolfe condition is also satisfied. According to Lemma A.1, the direction generated by the algorithm is a descent direction. Thus, the conditions of Lemma A.4 are satisfied, and the Zoutendijk condition holds.

There is a contradiction between (A.19) and the Zoutendijk condition, so we conclude that assumption (A.14) is not valid. Thus, we have $\lim_{k \rightarrow \infty} \|E_w^k\| = 0$.

Proving Theorem 3.3. According to (2.18), Lemma A.2, and Theorem 3.2, we have

$$\lim_{k \rightarrow \infty} \|w^{k+1} - w^k\| = \lim_{k \rightarrow \infty} \eta^k \|d^k\| \leq \lim_{k \rightarrow \infty} \varepsilon_1 \eta^k \|E_w^k\| = 0. \quad (\text{A.20})$$

Then, combining with Lemma A.5, we can obtain

$$\lim_{k \rightarrow \infty} w^k = w^*. \quad (\text{A.21})$$

We have completed the proof of Theorem 3.3.

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